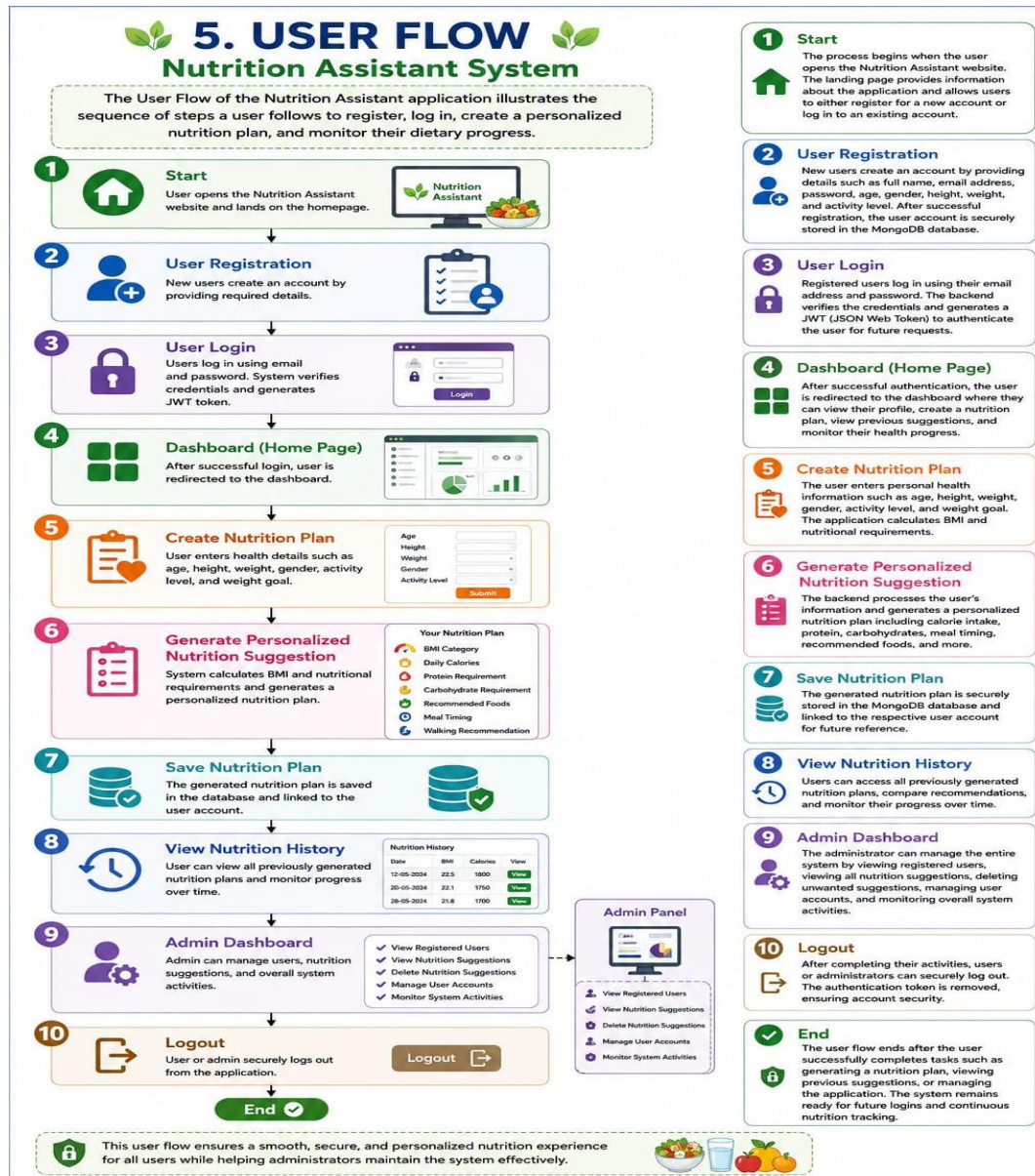


USER FLOW

The User Flow of the **Nutrition Assistant** application illustrates the sequence of steps a user follows to register, log in, create a personalized nutrition plan, and monitor their dietary progress. The system ensures a simple, secure, and user-friendly experience while delivering customized nutrition recommendations.



1. Start

The process begins when the user opens the Nutrition Assistant website. The landing page provides information about the application and allows users to either register for a new account or log in to an existing account.

2. User Registration

New users create an account by providing details such as:

- Full Name
- Email Address
- Password
- Age
- Gender
- Height
- Weight
- Activity Level

After successful registration, the user account is securely stored in the MongoDB database.

3. User Login

Registered users log in using their email address and password. The backend verifies the credentials and generates a JWT (JSON Web Token) to authenticate the user for future requests.

4. Dashboard (Home Page)

After successful authentication, the user is redirected to the dashboard where they can:

- View their profile
- Create a new nutrition plan
- View previous nutrition suggestions
- Monitor their health progress

5. Create Nutrition Plan

The user enters personal health information such as:

- Age
- Height
- Weight
- Gender
- Activity Level
- Weight Goal

The application calculates BMI and nutritional requirements based on the provided details.

6. Generate Personalized Nutrition Suggestion

The backend processes the user's information and generates a personalized nutrition plan that includes:

- BMI Category
- Daily Calorie Intake
- Protein Requirement
- Carbohydrate Requirement
- Recommended Foods
- Meal Timing
- Walking Recommendation
- Weight Management Suggestions

7. Save Nutrition Plan

The generated nutrition plan is securely stored in the MongoDB database and linked to the respective user account for future reference.

8. View Nutrition History

Users can access all previously generated nutrition plans, compare recommendations, and monitor their progress over time.

9. Admin Dashboard

The administrator can manage the entire system by:

- Viewing registered users

- Viewing all nutrition suggestions
- Deleting unwanted nutrition suggestions
- Managing user accounts
- Monitoring overall system activities

10. Logout

After completing their activities, users or administrators can securely log out. The authentication token is removed, ensuring account security.

End

The user flow ends after the user successfully completes tasks such as generating a nutrition plan, viewing previous suggestions, or managing the application. The system remains ready for future logins and continuous nutrition tracking.